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Positive Pressure Ventilation

Number: 0452

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Scope of Policy

This Clinical Policy Bulletin addresses positive pressure ventilation.

I. **Medical Necessity**

Aetna considers the following as medically necessary (unless otherwise specified) when criteria are met.

A. Noninvasive positive pressure ventilation (NPPV) with bilevel positive airway pressure (bilevel PAP, BIPAP) devices or a bilevel PAP device with a backup rate feature as durable medical equipment (DME) for members who have restrictive thoracic disorders (i.e., neuromuscular diseases or severe thoracic cage abnormalities), severe chronic obstructive pulmonary disease (COPD), central sleep apnea (CSA), complex sleep apnea (CompSA), hypoventilation syndrome, or obstructive sleep apnea (bilevel PAP without backup rate feature only), and who meet the medical necessity criteria for these conditions:

1. **Restrictive Thoracic Disorders**

a. Chronic obstructive pulmonary disease (COPD) does not contribute significantly to the member's pulmonary limitation; *and*
b. Member has a progressive neuromuscular disease (e.g., amyotrophic lateral sclerosis, etc.) or a severe thoracic cage abnormality (e.g., post-thoracoplasty for tuberculosis, etc.), *and*
c. Member has symptoms of sleep-associated hypoventilation (nocturnal hypoxemia), such as daytime hypersomnolence, excessive fatigue, dyspnea, morning headache, cognitive dysfunction, etc., *and*
d. Member has clinically significant hypoxemia, as indicated by *any* of the following:

i. An arterial blood gas PaCO2, done while awake and breathing the member's usual FIO2 (fractional inspired oxygen concentration), is greater than or equal to 45 mm Hg; *or*

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ii. Sleep oximetry demonstrates oxygen saturation less than or equal to 88 % for at least 5 minutes of nocturnal recording time (minimum recording time of 2 hours), done while breathing the member's usual FIO2; *or*
iii. For progressive neuromuscular disease only, maximal inspiratory pressures less than 60 cm H20 or forced vital capacity (FVC) less than 50 % predicted;

2. **Severe Chronic Obstructive Pulmonary Disease**

a. Member has symptoms of sleep-associated hypoventilation (nocturnal hypoxemia), such as daytime hypersomnolence, excessive fatigue, dyspnea, morning headache, cognitive dysfunction, etc.; *and*
b. Member has an arterial blood gas PaCO2 greater than or equal to 52 mm Hg; *and*
c. Sleep oximetry demonstrates oxygen saturation less than or equal to 88 % for at least 5 cumulative minutes of nocturnal recording time (minimum recording time of two hours), done while breathing oxygen at 2 liters per minute (LPM) or the member's prescribed FIO2, whichever is higher; *and*
d. Prior to initiating therapy, obstructive sleep apnea (OSA) (and treatment with continuous positive airway pressure (CPAP)) has been considered and ruled out. **Note:** Formal sleep testing is not required if there is sufficient information in the medical record to demonstrate that the member does not suffer from some form of sleep apnea (obstructive sleep apnea (OSA), CSA and/or CompSA) as the predominant cause of awake hypercapnia or nocturnal arterial oxygen desaturation.

If all of the above criteria for members with COPD are met, a bilevel PAP device without a backup rate feature will be considered medically necessary. A bilevel PAP device with a backup rate feature will only be considered medically necessary for COPD in *either* of the following situations below:

i. A bilevel PAP device with a backup rate feature will be considered medically necessary for COPD starting any time after initial use of a bilevel Pap device without a backup rate feature when the following criteria are met:

i. an arterial blood gas PaCO2, done while awake and breathing the member's prescribed FIO2, shows that the member's PaCO2 worsens greater than or equal to 7 mm Hg compared to the original result from criterion 2.b.(above); *and*
ii. a facility-based PSG demonstrates oxygen saturation less than or equal to 88% for greater than or equal to a cumulative 5 minutes of nocturnal recording time (minimum recording time of 2 hours) while using a bilevel PAP device without a backup rate feature, that is not caused by obstructive upper airway events – i.e., AHI less than 5 (see CPB 0004 - Obstructive Sleep Apnea in Adults for criteria for a bilevel PAP device for obstructive sleep apnea)*; or*

ii. A bilevel PAP device with a backup rate feature will be considered medically necessary for COPD when, after at least two months (60 days) of compliant use (an average of 4 hours use per 24 hour period) of a bilevel PAP device without a backup rate feature, *and*

i. an arterial blood gas PaCO2 is done while awake and breathing the member's prescribed FIO2, still remains greater than or equal to 52 mm Hg; *and*
ii. sleep oximetry while breathing with the bilevel PAP device without a backup rate feature, demonstrates oxygen saturation less than or equal to 88% for greater than or equal to a cumulative 5 minutes of nocturnal recording time (minimum recording time of 2 hours), done while

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breathing oxygen at 2 LPM or the member's prescribed FIO2 (whichever is higher);

3. **Central Sleep Apnea (CSA) or Complex Sleep Apnea (CompSA) (see Glossary and Appendix for definitions)**

   Prior to initiating therapy, a complete inpatient, attended polysomnogram must be performed documenting the following:

   a. The diagnosis of CSA or CompSA (see Appendix); *and*
   b. Significant improvement of the sleep-associated hypoventilation with the use of a bilevel PAP device with or without a backup rate feature on the settings that will be prescribed for initial use at home, while breathing the member's prescribed FIO2;

4. **Hypoventilation Syndrome**

   a. A bilevel device *without* a backup rate feature will be considered medically necessary for hypoventilation syndrome when criterion i *and* ii *plus* criterion iii *or* iv are met:

      i. An initial arterial blood gas PaCO2, done while awake and breathing the member's prescribed FIO2, is greater than or equal to 45 mm Hg.
      ii. Spirometry shows an FEV1/FVC greater than or equal to 70%. (Refer to Severe Chronic Obstructive Pulmonary Disease (above) for information about medical necessity for members with FEV1/FVC less than 70%.)
      iii. An arterial blood gas PaCO2, done during sleep or immediately upon awakening, and breathing the member's prescribed FIO2, shows the member's PaCO2 worsened greater than or equal to 7 mm Hg compared to the original result.
      iv. A facility-based polysomnography (PSG) or home sleep test (HST) demonstrates oxygen saturation less than or equal to 88% for greater than or equal to 5 minutes of nocturnal recording time (minimum recording time of 2 hours) that is not caused by obstructive upper airway events – i.e., AHI less than 5. (See CPB 0004 - Obstructive Sleep Apnea in Adults for for information about bilevel PAP devices for obstructive sleep apnea);

   b. Bilevel device *with* a backup rate feature will be considered medically necessary for hypoventilation syndrome when criterion i *and* ii *plus* criterion iii *or* iv are met:

      i. A medically necessary bilevel PAP device without a backup rate feature is being used.
      ii. Spirometry shows an FEV1/FVC greater than or equal to 70%. (Refer to Severe Chronic Obstructive Pulmonary Disease (above) for information about medical necessity for members with FEV1/FVC less than 70%).
      iii. An arterial blood gas PaCO2, done while awake, and breathing the member's prescribed FIO2, shows that the member's PaCO2 worsens greater than or equal to 7 mm Hg compared to the arterial blood gas result performed to qualify the member for the bilevel PAP device without a backup rate feature.
      iv. A facility-based PSG or HST demonstrates oxygen saturation less than or equal 88% for greater than or equal to 5 minutes of nocturnal recording time (minimum recording time of 2 hours) that is not caused by obstructive upper airway events – i.e., AHI less than 5 while using a bilevel device without a backup rate feature. (Refer to CPB 0004 - Obstructive Sleep Apnea in Adults for medical necessity criteria for bilevel PAP devices without backup rate for obstructive sleep apnea);

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5. **Obstructive Sleep Apnea (OSA)**

   A bilevel PAP device without a backup rate feature is considered medically necessary for persons with OSA if *both* of the following criteria are met:

   a. Member meets the criteria for CPAP, as set forth in the CPB 0004 - Obstructive Sleep Apnea in Adults or CPB 0752 - Obstructive Sleep Apnea in Children; *and*
   b. CPAP has been tried and proven ineffective or is not tolerated;

   A backup rate feature for a bilevel PAP device is of no proven value for the primary diagnosis of OSA and therefore will be considered experimental, investigational, or unproven.

6. **Tracheomalacia**

   Continuous positive airway pressure is considered medically necessary for the treatment of tracheomalacia;

7. **Respiratory Failure Following Surgery**

   Noninvasive positive pressure ventilation (NPPV) is considered medically necessary for postoperative hypoxemic respiratory failure that is refractory to or not suitable for oxygen;

8. **Continued Coverage Criteria Beyond the First Three Months of Therapy**

   Members should be re-evaluated after 2 to 3 months to evaluate their continued medical necessity for NPPV. For establishment of continued medical necessity beyond 3 months, the medical records should document that the member has been compliantly using the device (an average of 4 hours per 24-hour period), and that the member is benefiting from its use.

B. Either a heated or non-heated humidifier is considered medically necessary for use with NPPV.
C. Ventilator With Noninvasive Interfaces

Aetna considers ventilators with noninvasive interfaces medically necessary for severe neuromuscular diseases, thoracic restrictive diseases, and chronic respiratory failure consequent to chronic obstructive pulmonary disease where interruption or failure of respiratory support would lead to death. Aetna follows Centers for Medicare & Medicaid Services (CMS) policy on ventilators with noninvasive interfaces. A CMS National Coverage Determination states that ventilators are covered for the following conditions: “neuromuscular diseases, thoracic restrictive diseases, and chronic respiratory failure consequent to chronic obstructive pulmonary disease.” Each of these disease categories are comprised of conditions that can vary from severe and life-threatening to less serious forms. These disease groups may appear to overlap conditions described above for bilevel PAP devices but they are not overlapping. Choice of an appropriate device, i.e., a ventilator versus a bi-level PAP device is made based upon the severity of the condition. CMS distinguished the use of respiratory product types in a National Coverage Analysis Decision Memo stating that bilevel PAP devices are “distinguished from ventilation in a patient for whom interruption or failure of respiratory support leads to death.” The conditions described above for bilevel PAP devices are not life-threatening conditions where interruption of respiratory support would quickly lead to serious harm or death. These describe clinical conditions that require intermittent and relatively short durations of respiratory support. Thus, any type ventilator would not be considered medically necessary for any of the conditions described above for bilevel PAP devices even though the ventilator equipment may have the capability of operating in a bi-level PAP mode. Bi-level PAP devices are considered medically

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necessary in those clinical scenarios. Use of ventilators for the treatment of conditions described above for bilevel PAP devices is considered not medically necessary.

D. Multi-function Home Ventilator

Aetna considers an FDA-approved multi-function home ventilator (e.g., VOCSN, VOCSN VC, VOCSN VC Pro [Ventec Life Systems, Inc.]) medically necessary for members who meet *both* of the following criteria:

1. Member has documented diagnosis of a neuromuscular disease, thoracic restrictive disease, or chronic respiratory failure consequent to chronic obstructive pulmonary disease requiring positive pressure mechanical ventilation where interruption or failure of respiratory support would lead to death; *and*
2. Documentation supports the need for more than one additional function (i.e., oxygen concentrator, cough stimulator, suction pump, nebulization).

E. Second Respiratory Assist Device

Aetna considers a second invasive or non-invasive respiratory assist device medically necessary when it is required to serve a different purpose as determined by the member's medical needs. Examples (not all-inclusive) of situations in which multiple respiratory assist devices may be considered medically necessary are:

1. An individual requires one type of respiratory assist device (e.g., a negative pressure ventilator with a chest shell) for part of the day and needs a different type of respiratory assist device (e.g., positive pressure respiratory assist device with a nasal mask) during the rest of the day.
2. An individual who is confined to a wheelchair requires a respiratory assist device mounted on the wheelchair for use during the day and needs another respiratory assist device of the same type for use while in bed. Without both pieces of equipment, the individual may be prone to certain medical complications, may not be able to achieve certain appropriate medical outcomes, or may not be able to use the medical equipment effectively.

II. Experimental, Investigational, or Unproven

The following interventions are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

A. Noninvasive positive pressure ventilation (NPPV) for all other indications not listed in Section I, including the following (not an all-inclusive list):

- Acute lung injury
- As an alternative to endotracheal intubation following esophagectomy
- Asthma
- Bronchiolitis in infants and children
- Pneumonia
- Prevention of complications after pulmonary resection for lung cancer;

B. Single-Breath Tests for Determining Airway Closure Volume

Aetna considers single-breath nitrogen testing (also known as single-breath oxygen testing) experimental, investigational, or unproven because the value of this test in the management of persons with pulmonary disorders/diseases has not been established. Single-breath tests for determining airway closure volume that are performed using other tracer gases such as xenon, argon, or helium are also considered experimental, investigational, or unproven because of insufficient evidence in the peer-reviewed literature.

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C. Exsufflation belt (intermittent abdominal daytime pressure ventilator) is considered experimental, investigational, or unproven for pulmonary restrictive or pulmonary obstructive breathing (see also CPB 0067 - Chest Physiotherapy and Airway Clearance Devices).

III. Policy Limitations and Exclusions

A. A liner used in conjunction with a PAP mask is considered a comfort/convenience item.
B. Electrical generators to power respirators, bilevel PAP devices, etc. do not meet Aetna's definition of DME because they are not primarily medical in nature, and they are of use in the absence of illness and injury.

IV. Related Policies

- CPB 0004 - Obstructive Sleep Apnea in Adults
- CPB 0067 - Chest Physiotherapy and Airway Clearance Devices
- CPB 0752 - Obstructive Sleep Apnea in Children

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

CPT codes covered if selection criteria are met:

94002 - 94004	Ventilation assist and management, initiation of pressure or volume preset ventilators for assisted or controlled breathing
94660	Continuous positive airway pressure ventilation (CPAP), initiation and management

CPT codes not covered for indications listed in the CPB:

94726	Plethysmography for determination of lung volumes and, when performed, airway resistance

Other CPT codes related to the CPB:

82800 - 82810	Gases, blood
94760 - 94762	Noninvasive ear or pulse oximetry for oxygen saturation
95782	Polysomnography; younger than 6 years, sleep staging with 4 or more additional parameters of sleep, attended by a technologist
95783	younger than 6 years, sleep staging with 4 or more additional parameters of sleep, with initiation of continuous positive airway pressure therapy or bi-level ventilation, attended by a technologist

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Code	Code Description
95808 - 95811	Polysomnography

HCPCS codes covered if selection criteria are met:

A7027	Combination oral/nasal mask, used with continuous positive airway pressure device, each
A7028	Oral cushion for combination oral/nasal mask, replacement only, each
A7029	Nasal pillows for combination oral/nasal mask, replacement only, pair
A7030 - A7039	Full face mask, each, face mask interface replacement, each, replacement cushion for nasal application, each, replacement pillows, pair, nasal interface (mask or cannula type), with or without headstrap, headgear, chinstrap, tubing. filter, disposable or filter nondisposable, used with positive airway pressure device
A7044	Oral interface used with positive airway pressure device, each
A7045	Exhalation port with or without swivel used with accessories for positive airway devices, replacement only
A7046	Water chamber for humidifier, used with positive airway pressure device, replacement, each
E0465	Home ventilator, any type, used with invasive interface, (e.g., tracheostomy tube)
E0466	Home ventilator, any type, used with non-invasive interface, (e.g., mask, chest shell)
E0467	Home ventilator, multi-function respiratory device, also performs any or all of the additional functions of oxygen concentration, drug nebulization, aspiration, and cough stimulation, includes all accessories, components and supplies for all functions [FDA-approved multi-function home ventilator (e.g., VOCSN, VOCSN VC, VOCSN VC Pro [Ventec Life Systems])]
E0470	Respiratory assist device, bi-level pressure capability, without backup rate feature, used with non-invasive interface, e.g., nasal or facial mask (intermittent assist device with continuous positive airway pressure device)
E0471	Respiratory assist device, bi-level pressure capability, with back-up rate feature, used with noninvasive interface, e.g., nasal or facial mask (intermittent assist device with continuous positive airway pressure device) [*note - device with backup rate not covered for obstructive sleep apnea]
E0561	Humidifier, non-heated, used with positive airway pressure device
E0562	Humidifier, heated, used with positive airway pressure device
E0601	Continuous positive airway pressure (CPAP) device

HCPCS codes not covered for indications listed in the CPB:

A4468	Exsufflation belt, includes all supplies and accessories

ICD-10 codes covered if selection criteria are met:

B91	Sequelae of poliomyelitis
E66.2	Morbid (severe) obesity with alveolar hypoventilation [hypoventilation syndrome]
G12.0 - G12.9	spinal muscular atrophy and related syndromes
G14	Postpolio syndrome
G47.31	Primary central sleep apnea
G47.33	Obstructive sleep apnea (adult) (pediatric)
G47.35	Congenital central alveolar hypoventilation syndrome [hypoventilation syndrome]

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Code	Code Description
G47.36	Sleep related hypoventilation in conditions classified elsewhere [hypoventilation syndrome]
G47.37	Central sleep apnea in conditions classified elsewhere
G54.0 - G54.9, G55	Nerve root and plexus disorders
G70.00 - G73.7	Diseases of myoneural junction and muscle
G93.31 - G93.39	Postviral fatigue syndrome
J39.8	Other specified diseases of upper respiratory tract [tracheomalacia]
J40 - J44.9 J47.0 - J47.9	Chronic lower respiratory system diseases
J67.0 - J67.9	Respiratory diseases due to external agents
J80 J96.00 - J96.92 J98.4	Other diseases of the respiratory system
M41.00 - M41.9, M96.5	Scoliosis
M95.4	Acquired deformity of chest and rib [thoracic restrictive diseases]
Q32.0	Congenital tracheomalacia
Q67.8	Other congenital deformities of chest [thoracic restrictive diseases]
R06.00 - R06.9	Abnormalities of breathing
R09.02	Hypoxemia
R53.0 - R53.1 R53.81, R53.83	Other malaise and fatigue
Numerous options	Late effect of spinal cord injury or injury to nerve root(s), spinal plexus(es), and other nerves of trunk [Codes not listed due to expanded specificity]

ICD-10 codes not covered for indications listed in the CPB:

J10.00 - J18.9	Pneumonia and influenza
J21.0 - J21.9	Acute bronchiolitis [in infants and children]
J45.20 - J45.998	Asthma
S21.309A - S21.309S, S27.301A - S27.301S, S27.399A - S27.399S	Injury to lung

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Background

Positive pressure ventilation (PPV) is a form of respiratory therapy that uses a device to deliver oxygen into the lungs via positive pressure. PPV is delivered in one of two forms: noninvasive positive-pressure ventilation (via full-face mask, orofacial mask mouthpiece, or nasal mask/pillow) or invasive positive-pressure ventilation (via a tracheostomy tube when PPV therapy is used in a home setting).

Noninvasive positive-pressure ventilation (NPPV) is most commonly administered by a CPAP (continuous positive airway pressure) or BiPAP (bilevel positive airway pressure) machine. NPPV devices are often prescribed to prevent collapse of the oropharyngeal walls and the obstruction of airflow, such as in obstructive sleep apnea (OSA), and treat conditions that cause airflow blockage, such as in chronic obstructive pulmonary disease (COPD).

For NPPV indications, this policy focuses on the use of the bilevel PAP ventilator, and is based on Medicare policy and on the conclusions of a consensus conference on noninvasive positive pressure ventilation (NAMDRC, 1999).

According to Durable Medical Equipment Medicare Administrative Carrier (DME MAC) policy, noninvasive positive pressure respiratory assistance provided by a respiratory assist device is the administration of positive air pressure, using a nasal and/or oral mask interface which creates a seal, avoiding the use of more invasive airway access (e.g., tracheostomy). It may be applied to assist insufficient respiratory efforts in the treatment of conditions that may involve sleep-associated hypoventilation. It is to be distinguished from the invasive ventilation administered via a securely intubated airway, in a patient for whom interruption or failure of ventilatory support would lead to imminent demise of the patient.

NPPV for Restrictive Thoracic Diseases

A wide variety of restrictive thoracic diseases have been successfully treated with NPPV, including thoracic cage abnormalities (e.g., chest wall deformities, kyphoscoliosis, thoracoplasty, etc.) in addition to both rapidly and slowly progressive neuromuscular disorders (e.g., amyotrophic lateral sclerosis (ALS), neuropathies, myopathies, dystrophies, sequelae of polio, spinal cord injury, etc.). These conditions result in derangement of hypoventilation, and oxygen therapy alone is not only usually ineffective in relieving symptoms, but may also be dangerous and lead to a marked acceleration of carbon dioxide (CO_2) retention. Noninvasive positive pressure ventilation is generally not indicated for patients who can not cooperate with NPPV treatment or who need a protected airway to handle excessive secretions. (Patients who have impaired ability to protect the upper airway or excessive secretions are usually better managed with tracheostomy). The availability of a full face mask, however, has made it possible to use NPPV even in patients with significant bulbar weakness.

Indications for NPPV are based on symptoms attributable to nocturnal hypoventilation and objective findings of nocturnal de-saturation. The most common symptoms of chronic respiratory failure are associated with nocturnal sleep disruption, and include daytime hypersomnolence, excessive fatigue, morning headache, cognitive dysfunction, and even dyspnea. A consensus conference suggested that any $PaCO_2$ greater than or equal to 45 mm Hg or abnormal nocturnal oxygen de-saturation is a sufficient indication for NPPV. Clinically significant hypoxemia during sleep has been defined as an oxyhemoglobin saturation of less than or equal to 88 % for at least 5 minutes. This criterion for clinically significant nocturnal hypoxemia was favored because it is relatively simple to determine and is consistent with established guidelines for determination of hypoxemia for oxygen therapy.

For patients with progressive neuromuscular disorders, the consensus panel concluded that pulmonary function test results may be an additional indicator of nocturnal de-saturation. Most amyotrophic lateral sclerosis patients have a forced vital capacity (FVC) below 50 % predicted before either the physician or patient actually becomes aware of any respiratory system involvement. Other measurements like maximal inspiratory pressure with a magnitude less than 60 cm H_2O have been shown to be highly sensitive albeit less specific indicator of nocturnal de-saturation.

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What type of equipment and what specific ventilator settings should be chosen are controversial. Most studies of long-term NPPV for patients with neuromuscular disease have used volume-targeted rather than pressure-targeted devices. More recent reviews have cited the advantages of pressure-targeted devices for comfort and in their ability to compensate for leaks. Volume-targeted equipment may be favorable for patients simply because triggering mechanisms are more adjustable and pressure-targeted systems are not able to guarantee a minimum minute ventilation. The need for NPPV with a mandatory backup rate (e.g., Adaptive-Servoventilation), however, is more generally accepted because of the profound rapid eye movement (REM) de-saturation that often occurs in patients with respiratory muscle weakness.

Physician re-assessment of patient benefit and adherence to NPPV therapy should occur within 60 days of initiation of therapy. The specific methods used may be as simple as a patient interview to assess compliance but usually involve some assessment of awake arterial blood gas values and overnight oximetry while using the designated NPPV therapy.

NPPV for Chronic Obstructive Pulmonary Disease (COPD)

During the 1980s, investigators used negative-pressure ventilators, mainly of the tank or "wrap" type, to provide intermittent respiratory muscle rest in patients with severe COPD. However, a number of long-term controlled clinical studies showed negative-pressure ventilation to be of no benefit in pulmonary function, daytime gas exchange, or functional capability. Furthermore, patients tolerated negative-pressure ventilators poorly.

Clinical studies of NPPV in patients with COPD (e.g., chronic bronchitis, emphysema, bronchiectasis, cystic fibrosis, etc.) have shown that NPPV is better tolerated than negative-pressure ventilation. In addition, advantages of ease of administration and portability as well as the ability to eliminate obstructive sleep apnea (OSA) make NPPV the first choice of non-invasive modes.

Although the evidence is conflicting and far from definitive, the consensus conference concluded that patients with substantial daytime CO_2 retention, particularly those with nocturnal oxygen de-saturation, appear most apt to respond favorably to nocturnal NPPV. Patients with little or no CO_2 retention, regardless of the severity of airway obstruction, appear to gain little or no benefit from NPPV.

Struik et al (2014) stated that the effects of nocturnal NPPV in patients with stable COPD remain controversial. The Cochrane Airways group Register of Trials, MEDLINE, EMBASE and CINAHL were searched up to August 2012. Individual patient data from randomized controlled trials (RCTs) on NPPV outcomes were selected for 2 separate meta-analyses: the first with follow-up of 3 months and the second with 12 months of follow-up. Additionally, subgroup analyses within the NPPV group comparing inspiratory positive airway pressure (IPAP) levels, compliance and levels of hypercapnia on change in $PaCO_2$ after 3 months were performed. A total of 7 trials (245 patients) were included. All studies were considered of moderate to high quality. No significant difference was found between NPPV and control groups after 3 or 12 months of follow-up when looking at $PaCO_2$ and PaO_2, 6-minute walking distance, health-related quality-of-life, FEV1, FVC, maximal inspiratory pressure and sleep efficiency. Significant differences in change in $PaCO_2$ after 3 months were found for patients ventilated with IPAP levels of at least 18 cm H2O, for patients who used NPPV for at least 5 hours per night as well as for patients with baseline $PaCO_2$ of at least 55 mm Hg when compared to patients with lower IPAP levels, poorer compliance or lower levels of hypercapnia. The authors concluded that at present, there is insufficient evidence to support the application of routine NPPV in patients with stable COPD. However, higher IPAP levels, better compliance and higher baseline $PaCO_2$ seem to improve $PaCO_2$.

Wu et al (2022) stated that the evidence of long-term home non-invasive positive pressure ventilation (LTHNIPPV) in patients with stable hypercapnic COPD is controversial. In a meta-analysis, these investigators examined if a baseline level and reduction in PaCO2 were associated with the treatment effect of LTHNIPPV in these patients. A total of 6 electronic databases were searched from January 1980 until June 2020; RCTs comparing LTHNIPPV with control treatment were included. Two authors independently

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extracted data, assessed the study quality, and used the GRADE approach to evaluate evidence quality. The main outcome was mortality. A total of 19 studies involving 1,482 patients (LTHNIPPV, n = 730; control, n = 752) were included. LTHNIPPV significantly reduced mortality (risk ratio [RR] = 0.76; 95 % CI: 0.61 to 0.95; p = 0.02; I2 = 14 %), the frequency of hospital admissions, PaCO2, and improved PaO2 compared to control treatment. LTHNIPPV also relieved dyspnea and improved exercise capacity and health-related QOL (HR-QOL) but showed no significant benefit for improving FEV1% pred. Subgroup analysis revealed that the baseline level and reduction in PaCO2 were associated with decreased mortality (baseline PaCO2 of 55 mmHg or higher; RR = 0.69, p = 0.02; vs baseline PaCO2 of less than 55 mmHg; RR = 0.87, p = 0.32; and higher dPaCO2; RR = 0.42, p < 0.0001; versus lower dPaCO2; RR = 0.91, p = 0.38). The authors concluded that LTHNIPPV significantly reduced mortality. The baseline level and reduction in PaCO2 were associated with the treatment effect of LTHNIPPV in patients with stable hypercapnic COPD. Moreover, these investigators concluded from their results based on clinical trials that the mechanisms of the association between the reduction in PaCO2 and decreased mortality were still unclear; therefore, large-scale, multi-center RCTs are needed to confirm these findings.

The authors stated that this meta-analysis had several drawbacks. First, the prior screening time for the definition of stable hypercapnic varied, which may have affected these findings. To address this problem, subgroup analysis was carried out, and the results remained consistent. Moreover, there was no significant heterogeneity for the main outcome, mortality; thus, the key findings were credible. Second, the measurement tools and data types were different in some outcomes; therefore, not all results were combined and analyzed for some secondary outcomes, and the quality of these results was evaluated as low or very low. Third, the quality of the included studies was inconsistent; many studies had uncertain bias for selection, performance, and detection, and these risks of bias were considered when evaluating the quality of evidence.

NPPV for Other Respiratory Disorders Associated with Nocturnal Hypoventilation

A variety of other respiratory disorders have been shown to predispose patients to nocturnal hypoventilation. These include central (non-obstructive) sleep apnea, complex sleep apnea, and OSA.

Most reports covering the effect of non-invasive ventilation on hypoventilation have focused on neuromuscular/chest wall disorders and patients with COPD. In contrast, there are few reports on non-invasive ventilation in patients with other disorders leading to nocturnal hypoventilation that may be treated with NPPV. Furthermore, although there are many reports demonstrating the benefits of continuous positive airway pressure (CPAP) in patients with OSA, there are only limited data supporting the use of NPPV in these types of patients who fail to respond to CPAP therapy.

Based on available literature, certain general statements regarding indications for non-invasive positive pressure ventilation for other nocturnal hypoventilation syndromes can be made. Patients considered for this therapy should have the following: a disease known to cause hypoventilation; symptoms and signs of hypoventilation; failure to respond to first-line therapies in mild cases of hypoventilation (i.e., treatment of primary underlying disease with bronchodilators, respiratory stimulants, weight loss, supplemental oxygen, CPAP); or have moderate-to-severe hypoventilation.

A polysomnogram is required for diagnosis of sleep apnea. A CPAP trial is recommended if OSA is documented unless a previous CPAP trial was unsuccessful.

Potential side effects from NPPV include gastric distention, aspiration of gastric contents, conjunctivitis, facial abrasions from tight-fitting masks, hypotension, and mask dislocation leading to transient hypoxemia.

NPPV for Respiratory Failure after Extubation/Acute Hypoxemic Respiratory Failure

The need for re-intubation after extubation and discontinuation of mechanical ventilation is not uncommon and is associated with increased mortality. Noninvasive positive pressure ventilation has been suggested as a treatment for individuals with respiratory failure following extubation. In a multi-center, randomized, controlled trial (n = 221), Esteban et al (2004) examined the effect of NPPV on mortality in this clinical

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setting. These investigators concluded that NPPV does not prevent the need for re-intubation or reduce mortality in unselected patients who have respiratory failure following extubation. This is in agreement with the findings of Keenan et al (2002) who reported that the addition of NPPV to standard medical therapy does not improve outcome in heterogeneous groups of patients who develop respiratory distress during the first 48 hours after extubation.

Furthermore, in a recent review, Keenan et al (2004) evaluated the effect of NPPV on the rate of endotracheal intubation, intensive care unit and hospital length of stay, and mortality for patients with acute hypoxemic respiratory failure not due to cardiogenic pulmonary edema. The authors concluded that randomized trials suggest that patients with acute hypoxemic respiratory failure are less likely to require endotracheal intubation when NPPV is added to standard therapy. However, the effect on mortality is less clear, and the heterogeneity found among studies suggests that effectiveness varies among different populations. As a result, the literature does not support the routine use of NPPV in all patients with acute hypoxemic respiratory failure.

In a Cochrane review, Shah and colleagues (2005) stated that acute hypoxemic respiratory failure (AHRF) is an important cause of morbidity and mortality in children. Currently, positive pressure ventilation is the standard of care, although it is known to be associated with complications. Continuous negative extra-thoracic pressure ventilation (CNEP) or continuous positive airway pressure ventilation delivered via non-invasive approaches (Ni-CPAP) have demonstrated certain benefits in animal as well as uncontrolled human studies. These investigators evaluated the effectiveness of CNEP and Ni-CPAP in children with AHRF due to non-cardiogenic causes. They concluded that there is a lack of well-designed, controlled studies of non-invasive modes of respiratory support in pediatric patients with AHRF.

Rathi and colleagues (2017) described the characteristics and outcomes of critically ill cancer patients who received NPPV versus invasive mechanical ventilation as first-line therapy for AHRF. These researchers performed a retrospective cohort study of consecutive adult intensive care unit (ICU) cancer patients who received either conventional invasive mechanical ventilation or NPPV as first-line therapy for hypoxemic respiratory failure. Of the 1,614 patients included, the NPPV failure group had the greatest hospital length of stay, ICU length of stay, ICU mortality (71.3 %), and hospital mortality (79.5 %) as compared with the other 2 groups ($p < 0.0001$). The variables independently associated with NPPV failure included younger age (odds ratio [OR], 0.99; 95 % CI: 0.98 to 0.99; $p = 0.031$), non-Caucasian race (OR, 1.61; 95 % CI: 1.14 to 2.26; $p = 0.006$), presence of a hematologic malignancy (OR, 1.87; 95 % CI: 1.33 to 2.64; $p = 0.0003$), and a higher Sequential Organ Failure Assessment score (OR, 1.12; 95 % CI: 1.08 to 1.17; $p < 0.0001$). There was no difference in mortality when comparing early versus late intubation (less than or greater than 24 or 48 hours) for the NPPV failure group. The authors concluded that NPPV failure is an independent risk factor for ICU mortality, but NPPV patients who avoided intubation had the best outcomes compared with the other groups; early versus late intubation did not have a significant impact on outcomes.

NIPPV for Tracheomalacia

Tracheomalacia refers to softness or weakness of the trachea. It may occur in an isolated lesion or can be found in combination with other lesions that cause compression or damage of the airway. Tracheomalacia is usually benign, with symptoms due to airway obstruction. Conservative treatment is preferred in milder cases, since the outcome is usually favorable within the first 2 years of life. The clinical utility of non-specific treatments (e.g., anti-inflammatory agents, bronchodilators, antibiotics, physiotherapy) has not been proven by clinical trials. Airway surgery should be avoided, and non-invasive ventilation may be employed as a temporary measure. In very severe cases, aortopexy, trachostomy or stent placement are the preferred treatments (Fayon and Donato, 2010).

Davis et al (1998) stated that CPAP is used to minimize airway collapse in infants with tracheomalacia. Forced expiratory flows (FEFs) at functional residual capacity (FRC) increase with increasing CPAP in infants with tracheomalacia, and it has been suggested that CPAP prevents airway collapse by "stenting" the airway open. Since FEF is greater at higher than at lower lung volumes, these researchers evaluated whether the increase in flow measured at FRC (V FRC) with CPAP could be explained by the increase in FRC with CPAP. They measured full FEF-volume curves at CPAP levels of 0, 4, and 8 cm H2O in 6 infants with

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tracheomalacia and 5 healthy control infants. In both groups of infants, FVC did not change with CPAP; however, inspiratory capacity (IC) decreased and thus FRC increased with increasing CPAP. FEFs at FRC increased with increasing levels of CPAP; however, the FEFs at 50 % and 75 % of expired volume were not different for the 3 levels of CPAP for both groups of infants. These findings indicated that FEFs measured at the same lung volumes did not differ for the different levels of CPAP indicates that CPAP affects forced flows primarily by increasing lung volume.

In a prospective, randomized, controlled study, Essouri et al (2005) evaluated the efficacy of CPAP ventilation in infants with severe upper airway obstruction and compared CPAP to bilevel positive airway pressure (BIPAP) ventilation. A total of 10 infants (median age of 9.5 months, range of 3 to 18) with laryngomalacia (n = 5), tracheomalacia (n = 3), tracheal hypoplasia (n = 1), and Pierre Robin syndrome (n = 1) were included in this analysis. Breathing pattern and respiratory effort were measured by esophageal and trans-diaphragmatic pressure monitoring during spontaneous breathing, with or without CPAP and BIPAP ventilation. Median respiratory rate decreased from 45 breaths/min (range of 24 to 84) during spontaneous breathing to 29 (range of 18 to 60) during CPAP ventilation. All indices of respiratory effort decreased significantly during CPAP ventilation compared to un-assisted spontaneous breathing (median, range): esophageal pressure swing from 28 to 10 cm H(2)O (13 to 76 to 7 to 28), esophageal pressure time product from 695 to 143 cm H(2)O/s per minute (264 to 1,417 to 98 to 469), diaphragmatic pressure time product from 845 to 195 cm H(2)O/s per minute (264 to 1,417 to 159 to 1,183). During BIPAP ventilation a similar decrease in respiratory effort was observed but with patient-ventilator asynchrony in all patients. The authors concluded that this short-term study showed that non-invasive CPAP and BIPAP ventilation are associated with a significant and comparable decrease in respiratory effort in infants with upper airway obstruction. However, BIPAP ventilation was associated with patient-ventilator asynchrony.

Masters and Chang (2005) noted that tracheomalacia, a disorder of the large airways where the trachea is deformed or malformed during respiration is commonly seen in tertiary pediatric practice. It is associated with a wide spectrum of respiratory symptoms from life-threatening recurrent apnea to common respiratory symptoms such as chronic cough and wheeze. Current practice following diagnosis of tracheomalacia include medical approaches aimed at reducing associated symptoms of tracheomalacia, ventilation modalities of CPAP and BiPAP and, surgical approaches aimed at improving the caliber of the airway (airway stenting, aortopexy, tracheopexy). In a Cochrane review, these investigators evaluated the efficacy of medical and surgical therapies for children with intrinsic (primary) tracheomalacia. The Cochrane Central Register of Controlled Trials (CENTRAL), the Cochrane Airways Group Specialized Register, MEDLINE and EMBASE databases were searched by the Cochrane Airways Group. The latest searches were performed in February 2005. All randomized controlled trials of therapies related to symptoms associated with primary or intrinsic tracheomalacia were included in this analysis. Results of searches were reviewed against pre-determined criteria for inclusion. No eligible trials were identified and thus no data were available for analysis. No randomized controlled trials (RCTs) that examined therapies for intrinsic tracheomalacia were found. Eight of the more recent (last 11 years) non-RCTs reported a benefit from the various surgical interventions. The success was however not universal and in some studies severe adverse events occurred. The authors concluded that there is currently an absence of evidence to support any of the therapies currently utilized for management of intrinsic tracheomalacia. It is unlikely that any RCT on surgically based management will ever be available for children with severe life-threatening illness associated with tracheomalacia. For those with less severe disease, RCTs are clearly needed. Outcomes of these RCTs should include measurements of the trachea and physiological outcomes in addition to clinical outcomes.

An UpToDate review on "Tracheomalacia and tracheobronchomalacia in adults" (Ernst et al, 2012) states that "[c]ontinuous positive airway pressure (CPAP) can maintain an open airway and facilitate secretion drainage. This is often initiated in the hospital during an acute illness. The patient initially receives continuous CPAP and is gradually transitioned to intermittent CPAP as tolerated. Patients may use intermittent CPAP as long-term therapy. However, CPAP does not appear to have a long-term impact on dyspnea or cough. Positive airway pressure other than CPAP (e.g., bilevel positive airway pressure) may be used instead if hypercapnic respiratory failure exists".

Also an eMedicine article on "Tracheomalacia Treatment & Management" (Schwartz) states that " [s]upportive therapy is provided to most infants. Most respond to conservative management, consisting of

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humidified air, chest physical therapy, slow and careful feedings, and control of infection and secretions with antibiotics. The use of continuous positive airway pressure (CPAP) has been recommended in patients having respiratory distress and may be successful in patients requiring a short-term intervention as the disorder spontaneously resolves.

NPPV for Other Conditions

Chermont et al (2009) examined the effects of CPAP on exercise tolerance in outpatients with chronic heart failure (CHF). Following a double-blind, randomized, cross-over, and placebo-controlled protocol, 12 patients with CHF (8 males and 4 females; age of 54 +/- 12 years; body mass index 27.3 +/- 1.8 kg/m2, New York Heart Association Class II, III) underwent CPAP via nasal mask for 30 mins in a recumbent position. Mask pressure was 3 cm H2O for 10 mins, followed by individual progression up to 4 to 6 cm H2O, whereas placebo was fixed 0 to 1 cm H2O. A 6-min walk test was performed after placebo and CPAP. Continuous positive airway pressure decreased the resting heart rate (pre = 80 +/- 17 bpm; post = 71 +/- 15 bpm; p = 0.001) and mean arterial pressure (pre = 103 +/- 14 mm Hg; post = 97 +/- 13 mm Hg; p = 0.008). During exercise test, CPAP increased the distance covered (CPAP: 538 +/- 78 m; placebo: 479 +/- 83 m; p < 0.001) and the peak heart rate (CPAP: 98 +/- 17 bpm; placebo: 89 +/- 12 bpm; p = 0.049) but did not change the peak mean arterial pressure (p = 0.161). The authors concluded that non-invasive ventilation with CPAP increased exercise tolerance in patients with stable CHF. They stated that future clinical trials should investigate if this effect is associated with improved clinical outcome.

Keenan and Mehta (2009) summarized randomized controlled trials (RCTs) on non-invasive ventilation (NIV) for acute respiratory failure (ARF). These researchers conducted an extensive literature search and selected RCTs from that search. The results were presented primarily by etiology of respiratory failure, but they also included a short section on NIV for ARF in immunocompromised patients. The latter studies included patients with various etiologies of respiratory failure but with the common co-morbidity of immunocompromization. Most of the RCTs have studied NIV for exacerbation of COPD or cardiogenic pulmonary edema. In general, the RCTs have been small and used endotracheal intubation or NIV failure rate as primary outcomes. These investigators concluded that NIV for ARF is supported by strong evidence from patients with COPD, but there is only weak support for NIV in other patient groups, such as immunocompromised patients. For other groups, such as patients with asthma, pneumonia, or acute lung injury, RCT-level evidence is lacking or does not suggest benefit.

In a single-center RCT, Menesese et al (2011) examined if early nasal intermittent positive-pressure ventilation (NIPPV) compared with nasal continuous positive airway pressure (NCPAP) decreases the need for mechanical ventilation in infants with respiratory distress syndrome. Infants (gestational ages of 26 to 33/7 weeks) with respiratory distress syndrome were randomly assigned to receive early NIPPV or NCPAP. Surfactant was administered as rescue therapy. The primary outcome was the need for mechanical ventilation within the first 72 hours of life. A total of 200 infants, 100 in each arm, were randomly assigned. Rates of the primary outcome did not differ significantly between the NIPPV (25 %) and NCPAP (34 %) groups (relative risk [RR]: 0.71 [95 % confidence interval (CI): 0.48 to 1.14]). In post-hoc analysis, from 24 to 72 hours of life, significantly more infants in the NIPPV group remained extubated compared with those in the NCPAP groups (10 versus 22 %; RR: 0.45 [95 % CI: 0.22 to 0.91]). This difference was also noted in the group of infants who received surfactant therapy, NIPPV (10.9 %), and NCPAP (27.1 %) (RR: 0.40 [95 % CI: 0.18 to 0.86]). The authors concluded that early NIPPV did not decrease the need for mechanical ventilation compared with NCPAP, overall, in the first 72 hours of life. Moreover, they stated that further studies are needed to to evaluate the potential benefits of non-invasive ventilation, especially for the most vulnerable or preterm infants.

Najaf-Zadeh and Leclerc (2011) examined the effectiveness of NPPV in children less than 1 month of age with ARF due to different conditions. These researchers noted that mechanical respiratory support is a critical intervention in many cases of ARF. In recent years, NPPV has been proposed as a valuable alternative to invasive mechanical ventilation (IMV) in this acute setting. Recent physiological studies have demonstrated beneficial effects of NPPV in children with ARF. Several pediatric clinical studies, the majority of which were non-controlled or case series and of small size, have suggested the effectiveness of NPPV in the treatment of ARF due to acute airway (upper or lower) obstruction or certain primary

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parenchymal lung disease, and in specific circumstances, such as post-operative or post-extubation ARF, immunocompromised patients with ARF, or as a means to facilitate extubation. Noninvasive positive pressure ventilation was well-tolerated with rare major complications and was associated with improved gas exchange, decreased work of breathing, and endo-tracheal intubation avoidance in 22 to 100 % of patients. High FiO2 needs or high PaCO2 level on admission or within the first hours after starting NPPV appeared to be the best independent predictive factors for the NPPV failure in children with ARF. However, many important issues, such as the identification of the patient, the right time for NPPV application, and the appropriate setting, are still lacking. The authors concluded that further RCTs that address these issues in children with ARF are recommended.

Acute Respiratory Failure Following Upper Abdominal Surgery

In a Cochrane review, Faria and colleagues (2015) compared the safety and effectiveness and safety of NPPV versus standard oxygen therapy in the treatment of acute respiratory failure after upper abdominal surgery. These researchers evaluated the safety and effectiveness of NPPV (i.e., CPAP or bi-level NPPV) in reducing mortality and the rate of tracheal intubation in adults with acute respiratory failure after upper abdominal surgery, compared to standard therapy (oxygen therapy), and assessed changes in arterial blood gas levels, hospital and iICU length of stay, gastric insufflation, and anastomotic leakage. The date of the last search was May 12, 2015. These investigators searched the following databases: the Cochrane Handbook for Systematic Reviews of Interventions (CENTRAL) (2015, Issue 5), MEDLINE (Ovid SP, 1966 to May 2015), EMBASE (Ovid SP, 1974 to May 2015); the physiotherapy evidence database (PEDro) (1999 to May 2015); the Cumulative Index to Nursing and Allied Health Literature (CINAHL, EBSCOhost, 1982 to May 2015), and LILACS (BIREME, 1986 to May 2015). They reviewed reference lists of included studies and contacted experts. They also searched grey literature sources, and checked databases of ongoing trials such as www.controlled-trials.com/ and www.trialscentral.org/. They did not apply language restrictions. The authors selected RCTs or quasi-RCTs involving adults with acute respiratory failure after upper abdominal surgery who were treated with CPAP or bi-level NPPV with, or without, drug therapy as standard medical care, compared to adults treated with oxygen therapy with, or without, standard medical care. Two authors independently selected and abstracted data from eligible studies using a standardized form. They evaluated study quality by assessing allocation concealment; random sequence generation; incomplete outcome data; blinding of participants, personnel, and outcome assessors; selective reporting; and adherence to the intention-to-treat (ITT) principle. These investigators included 2 trials involving 269 participants. The participants were mostly men (67 %); the mean age was 65 years. The trials were conducted in China and Italy (1 was a multi-center trial). Both trials included adults with acute respiratory failure after upper abdominal surgery. These researchers judged both trials at high risk of bias. Compared to oxygen therapy, CPAP or bi-level NPPV may reduce the rate of tracheal intubation (RR 0.25; 95 % CI: 0.08 to 0.83; low quality evidence) with a number needed to treat for an additional beneficial outcome of 11. There was very low quality evidence that the intervention may also reduce ICU length of stay (mean difference (MD) -1.84 days; 95 % CI: -3.53 to -0.15). They found no differences for mortality (low quality evidence) and hospital length of stay. There was insufficient evidence to be certain that CPAP or NPPV had an effect on anastomotic leakage, pneumonia-related complications, and sepsis or infections. Findings from 1 trial of 60 participants suggested that bi-level NPPV, compared to oxygen therapy, may improve blood gas levels and blood pH 1 hour after the intervention (partial pressure of arterial oxygen (PaO2): MD 22.5 mm Hg; 95 % CI: 17.19 to 27.81; pH: MD 0.06; 95 % CI: 0.01 to 0.11; partial pressure of arterial carbon dioxide (PaCO2) levels (MD -9.8 mm Hg; 95 % CI: -14.07 to -5.53). The trials included in this systematic review did not present data on the following outcomes that these researchers intended to assess: gastric insufflation, fistulae, pneumothorax, bleeding, skin breakdown, eye irritation, sinus congestion, oronasal drying, and patient-ventilator asynchrony. The authors concluded that the findings of this review indicated that CPAP or bi-level NPPV is a safe and effective intervention for the treatment of adults with acute respiratory failure after upper abdominal surgery. However, they stated that based on the Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology, the quality of the evidence was low or very low; and more good quality studies are needed to confirm these findings.

An UpToDate review of management of postoperative pulmonary complications (Conde and Adams, 2016) stated that, although noninvasive ventilation is not routinely applied as a strategy to prevent postoperative

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respiratory failure, it is typically used as a secondary intervention for the treatment of hypoxemic respiratory failure that is refractory to or not suitable for low-flow or high-flow oxygen. The authors stated that the best support for this comes from a study of 293 patients with respiratory failure following abdominal surggery (citing Jaber, et al., 2016), which found that the use of noninvasive ventilation compared to standard oxygen therapy reduced the risk of tracheal intubation within 7 days.

Obesity Hypoventilation Syndrome

Masa and associates (2015) stated that the incidence of obesity hypoventilation syndrome (OHS) may be increasing in parallel with the present obesity epidemic. Despite extensive NIV and CPAP use in patients with OHS, information regarding effectiveness is limited. These researchers performed a large, multi-center RCT to determine the comparative effectiveness of NIV, CPAP, and lifestyle modification (control group) using daytime PaCO2 as the main outcome measure. Sequentially screened patients with OHS with severe sleep apnea were randomized into the above-mentioned groups for a 2-month follow-up. Arterial blood gas parameters, clinical symptoms, health-related quality-of-life assessments, polysomnography, spirometry, 6-minute-walk distance, drop-outs, compliance, and side effects were evaluated. Statistical analysis was performed using intention-to-treat analysis, although adjustments for CPAP and NIV compliance were also analyzed. A total of 351 patients were selected, and 221 were randomized; NIV yielded the greatest improvement in PaCO2 and bicarbonate, with significant differences relative to the control group, but not relative to the CPAP group. In the CPAP group, PaCO2 improvement was significantly different than in the control group only after CPAP compliance adjustment. Additionally, clinical symptoms and polysomnographic parameters improved similarly with NIV and CPAP relative to the control. However, some health-related quality-of-life assessments, the spirometry, and 6-minute-walk distance results improved more with NIV than with CPAP. Drop-outs were similar between groups, and compliance and secondary effects were similar between NIV and CPAP. The authors concluded that NIV and CPAP were more effective than lifestyle modification in improving clinical symptoms and polysomnographic parameters, although NIV yielded better respiratory functional improvements than did CPAP. Moreover, they stated that long-term studies must demonstrate whether this functional improvement has relevant implications.

An UpToDate review of obesity hypoventilation syndrome (Martin, 2016) stated that nocturnal noninvasive positive airway pressure is first-line treatment for obesity hypoventilation syndrome, regardless of whether or not the patient has a coexisting sleep-related breathing disorder (obstructive sleep apnea). "It is indicated for all patients with OHS and should NOT be delayed while the patient tries to lose weight." Patients with obesity hypoventilation syndrome alone are generally managed with bilevel positive airway pressure (BPAP). In contrast, patients with obesity hypoventilation syndrome and coexisting obstructive sleep apnea are usually managed initially with continuous positive airway pressure (CPAP) and then changed to BPAP if the CPAP is insufficient.

NPPV Following Esophagectomy

Raman et al (2015) noted that respiratory complications occur in 20 % to 65 % of patients who have undergone esophagectomy. While NPPV is associated with fewer complications than endotracheal intubation (ET), it is relatively contraindicated after esophagectomy due to potential injury to the anastomosis. These researchers created ex-vivo and in-vivo pig models to determine the pressure tolerance of an esophagectomy anastomosis and compared it to esophageal pressure during NPPV. These investigators created a stapled side-to-side, functional end-to-end esophago-gastric anastomosis. With continuous intraluminal pressure monitoring, they progressively insufflated the anastomosis with a syringe until an anastomotic leak was detected, and recorded the maximum pressure before leakage. These researchers performed this experiment in 10 esophageal specimens and 10 live pigs. They then applied a laryngeal mask airway (LMA) to 5 live pigs and measured the pressure in the proximal esophagus with increasing ventilatory pressures. The perforation was always at the anastomosis. The ex-vivo and in-vivo anastomoses tolerated a mean of 101 ± 44 cm H2O and 84 ± 38 cm H2O before leak, respectively. There was no significant difference between the pressure thresholds of ex-vivo and in-vivo anastomoses ($p = 0.51$). When 20, 30, and 40 cm H2O of positive pressure via LMA were delivered, the esophagus sensed 5 ± 4 cm H2O (25 %), 11 ± 11 cm H2O (37 %), and 15 ± 9 cm H2O (38 %), respectively. The authors concluded that the

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findings from their ovine model suggested that an esophagectomy anastomosis can tolerate a considerably higher pressure than is transmitted to the esophagus during NPPV. They stated that NPPV may be a safe alternative to ET after esophagectomy. These preliminary findings need to be validated in well-designed clinical trials.

Adaptive-Servoventilation

Adaptive servo-ventilation (ASV), a bilevel PAP system with a backup rate feature, uses an automatic, minute ventilation-targeted device (VPAP Adapt, ResMed, Poway, CA) that performs breath-to-breath analysis and adjusts its settings accordingly. Depending on breathing effort, the device will automatically adjust the amount of airflow it delivers in order to maintain a steady minute ventilation. Most studies on the use of ASV have investigated its use for heart failure patients with central apnea or Cheyne-Stokes respiration (Teschler et al, 2001; Pepperell et al, 2003; Topfer et al, 2004; Pepin et al, 2006; Kasai et al, 2006; Zhang et al, 2006).

Banno et al (2006) evaluated 3 patients with idiopathic Cheyne-Stokes breathing (CSB) and examined the feasibility of using ASV to treat them. The patients had a periodic breathing pattern resembling CSB. During polysomnography, the abnormal breathing pattern was present while patients were both awake and asleep. The patients were first tested on CPAP and/or oxygen; however they did not respond well to either of these treatments. They were then assessed on ASV. The mean abnormal breathing events index decreased from 35.2 to 3.5 per hour of sleep on ASV. There was a significant reduction in the mean number of arousals caused by abnormal breathing events: from 18.5 to 1.1 per hour of sleep. After 6 to 12 months of using ASV, the patients had maintained significant improvement in subjective daytime alertness and mood. The authors concluded that a trial of ASV for patients with idiopathic CSB is recommended if they do not have improvement in sleep respiration or daytime performance on CPAP and/or oxygen.

Morrell et al (2007) stated that hypercapnic cerebral vascular reactivity (HCVR) is reduced in patients with CHF and sleep-disordered breathing (SDB) and that this may be associated with an increased risk of stroke. These researchers tested the hypothesis that reversal of SDB in CHF patients using ASV would increase morning HCVR. A total of 10 CHF patients with SDB, predominantly OSA, were included in this study. The HCVR was measured from the change in middle cerebral artery velocity, using pulsed Doppler ultrasound. Hypercapnic cerebral vascular reactivity was determined during the evening (before) and morning (after) 1 night of sleep on ASV and 1 night of spontaneous sleep (control). Compared with the control situation, ASV decreased the apnea-hypopnea index (AHI) (group mean +/- SEM, control: 48 +/- 12, ASV: 4 +/- 1 events per hour). Hypercapnic cerebral vascular reactivity was 23 % lower in the morning, compared with the evening, on the control night (evening: 1.3 +/- 0.2, morning: 1.0 +/- 0.2 cm/sec per mm Hg, $p < 0.05$) and 27 % lower following the ASV night (evening: 1.5 +/- 0.2, morning: 1.1 +/- 0.2 cm/sec per mm Hg, $p < 0.05$). The effect of ASV on the evening-to-morning reduction in HCVR was not significant, compared with the control night (0.02 cm/sec per mm Hg, 95 % confidence interval: -0.28 to 0.32; $p = 0.89$). The authors concluded that in CHF patients with SDB, HCVR was reduced in the morning compared with the evening. However, removal of SDB for 1 night did not reverse the reduced HCVR. The relatively low morning HCVR could be linked with an increased risk of stroke.

Morgenthaler et al (2007) compared the efficacy of ASV versus NPPV for central, mixed, and complex sleep apnea syndromes in a prospective randomized cross-over clinical trial. A total of 21 patients (6 with central sleep apnea/Cheyne-Stokes respiration, 6 with predominantly mixed apneas, and 9 with complex sleep apnea) with initial diagnostic AHI +/- standard deviation 51.9 +/- 22.8/hr and RAI 45.5 less than or equal to 26.5/hr completed the study. Following optimal titration with CPAP ($n = 15$), disturbed breathing and disturbed sleep remained high with mean AHI = 34.3 +/- 25.7 and RAI = 32.1 +/- 29.7. AHI and RAI were markedly reduced with both NPPV (6.2 +/- 7.6 and 6.4 +/- 8.2) and ASV (0.8 +/- 2.4 and 2.4 +/- 4.5). Treatment AHI and RAI were both significantly lower using ASV ($p < 0.01$). The authors concluded that in patients with central sleep apnea/Cheyne-Stokes respiration, mixed apneas, and complex sleep apnea, both NPPV and ASV are effective in normalizing breathing and sleep parameters, and that ASV does so more effectively than NPPV in these types of patients.

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Hastings et al (2010) assessed the use of ASV in CHF patients with all types of sleep apnea. A total of 11 male patients with stable CHF and sleep apnea (AHI greater than 15 events/hr) were treated with 6 months optimized ASV and compared to 8 patients not receiving ASV. At baseline, both groups were comparable for New York Heart Association class, left ventricular ejection fraction (LVEF), plasma brain natriuretic peptide (BNP) concentrations and AHI. All patients were receiving optimal medical therapy. At 6 months, the authors reported that ASV significantly reduced AHI with improvement in LVEF and aspects of quality of life.

Single Breath Nitrogen Test

The single breath nitrogen test (SBNT) is a pulmonary function test that provides information on the evenness of distribution of ventilation and on closing volume. The test utilizes resident nitrogen (N_2) in the lung as the tracer gas, and a single inhalation of 100 % oxygen to cause a change in the N_2 concentration in the lungs. It is performed by having the subject breathe air normally through a mouthpiece, and after a single vital capacity inspiration of 100 % O_2, expire slowly and smoothly to residual volume. Expired N_2 concentration is then plotted against expired volume (single breath nitrogen washout curve). From this, information about the distribution of ventilation can be obtained. Similar measurements may be made using other tracer gases such as xenon, argon, or helium.

There are usually 4 phases to the single breath nitrogen washout curve – phase I represents dead space gas containing zero N_2; phase II is mixed dead space and alveolar gas; phase III is gas from the alveoli; and phase IV represents a sharp increase in N_2 concentration. In normal persons in whom the alveoli empty synchronously, phase III shows a plateau during which N_2 concentration rises only slowly. The slope of phase III (change in N_2 concentration per 500 ml of expired air) should be less than 1.5 %. The lung volume at which phase III changes to phase IV is closing volume (the volume at which closure of airways occur in the lower part of the lungs). An increase in closing volume, especially when it is larger than functional residual volume, indicates premature closure of intra-pulmonary airways as a result of the narrowing of small airways or reduced elastic recoil.

It was thought that the SBNT might detect chronic airway disease before it is clinically apparent. However, it has not been demonstrated conclusively to be more sensitive than other tests.

Most patients with established disease and an abnormal slope of phase III do not produce single breath tests from which closing volumes can be measured. The American Thoracic Society (ATS) Standards for the Diagnosis and Care of Patients with Chronic Pulmonary Disease (1995) notes that small airways (i.e., less than 2 mm in diameter) are important sites of airflow obstruction, and that the relative contribution of peripheral airway disease and loss of elastic recoil from emphysema may vary. However, the ATS Standards states that indices such as the closing capacity and the slope of the alveolar plateau derived from a SBNT are unable to identify individuals susceptible to chronic airway obstruction with cigarette smoke exposure. The ATS Standards notes that tests reflecting emphysema (e.g., single-breath diffusing capacity, functional residue capacity, total lung capacity) predict survival in a relatively minor way.

Fraser et al (1999) concluded that the SBNT has not been shown to be useful in identifying patients at risk for developing COPD. Fraser explained that, although epidemiological studies have demonstrated that the results of the SBNT is abnormal in many asymptomatic smokers, there is controversy regarding the value of this measurement, as it appears that this test may not offer advantages over simple spirometry in detecting the progression of airflow obstruction. Fraser et al explained that one reason SBNT has been less discriminating than was originally hoped in identifying smokers at risk for the development of progressive disease is the marked inter-subject and intra-subject variability in test results. In addition, these investigators noted that it has not been convincingly shown that the rate of decrease in forced expiratory flow in smokers correlates with abnormalities in small airway function.

A number of empirical studies have documented the limited clinical value of SBNT. Teculescuet et al (1988) noted that the SBNT did not detect any effect of involuntary smoking in a limited sample of children. Vestbo

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and Rasmussen (1990) reported that indices of the SBNT (e.g., closing volume, closing capacity, and slope of phase III) have no predictive value concerning overall mortality and cancer incidence. Vestbo et al (1990) concluded that in a random population sample indices from only one SBNT do not provide prognostic information concerning hospitalization in addition to that provided by forced expiratory volume in 1 sec (FEV1). Viegi et al (1988) stated that the place of SBNT in large scale epidemiologic testing has not been justified. Detels et al (1982) reported that the SBNT yielded less specific or different information than spirometry, the flow-volume curve, and the ratio of FEV1 to forced vital capacity (FVC) in identifying abnormal lung function. Reporting on SBNT and FEV1 in a cohort of individuals followed over a 9- to 11-year period, Vollmer et al (1990) concluded that SBNT variables are less reproducible than FEV1. Dahlqvist (1995) reported on the results of an 8-year correlational study involving 24 healthy subjects, and concluded that the "prognostic value of an abnormal single-breath nitrogen wash-out seems to be limited" in predicting an accelerated decline in FEV1. Moreover, Bourgkard et al (1997) reported that subjects with dust exposure and roentgenologic pneumoconiosis nodulation were unable to adequately perform SBNT; however, these subjects were able to performed spirometry satisfactorily. Thus, the SBNT has not been proven to be useful in detecting early lung dysfunction and selecting persons at risk for appropriate measures to prevent progression to advanced disease.

NPPV for Heart Failure

Momomura and associates (2015) noted that adaptive servo-ventilation (ASV) therapy using an innovative ventilator, originally developed to treat SDB, is a novel modality of NPPV and is gaining acceptance among Japanese cardiologists in expectation of its applicability to treat patients with chronic heart failure (CHF) based on its acute beneficial hemodynamic effects. These researchers conducted a multi-center, retrospective, real-world observational study in 115 Japanese patients with CHF, who had undergone home ASV therapy for the first time from January through December 2009, to examine their profile and the effects on their symptoms and hemodynamics. Medical records were used to investigate New York Heart Association (NYHA) class, echocardiographic parameters including left ventricular ejection fraction (LVEF), cardio-thoracic ratio (CTR), brain natriuretic peptide (BNP), and other variables. Most of the patients were categorized to NYHA classes II (44.4 %) and III (40.7 %); SDB severity was not determined in 44 patients, and SDB was not detected or was mild in 27 patients. In at least 71 patients (61.7 %), therefore, ASV therapy was not applied for the treatment of SDB. Chronic heart failure was more severe, i.e., greater NYHA class, lower LVEF, and higher CTR, in 87 ASV-continued patients (75.7 %) than in 28 ASV-discontinued patients (24.3 %). However, SDB severity was not related to continuity of ASV. The combined proportion of NYHA classes III and IV ($p = 0.012$) and LVEF ($p = 0.009$) improved significantly after ASV therapy; CTR and BNP did not improve significantly after ASV therapy but showed significant beneficial changes in their time-course analysis ($p < 0.05$, respectively). Improvements in LVEF and NYHA class after ASV therapy were not influenced by SDB severity at onset. The authors concluded that the findings of the present study suggested that ASV therapy would improve the symptoms and hemodynamics of CHF patients, regardless of SDB severity. Moreover, they stated that a randomized clinical study is needed to verify these effects.

Jiang and colleagues (2016) stated that NPPV alleviates SDB and it may improve cardiac function in patients with SDB. Because large RCTs directly evaluating the impact of NPPV on cardiac function are lacking, these researchers conducted a meta-analysis of published data on effectiveness of NPPV in improving cardiac function in patients with CHF regardless of SDB presence. Controlled trials were identified in PubMed, OVID, and Embase databases. Both fixed and randomized models were used in meta-analysis with primary outcomes of LVEF. A total of 19 studies were included with a total of 843 patients. Compared to standard medical treatment (SMT) plus sham-NPPV or SMT only, NPPV plus SMT was associated with improvement in LVEF (weighted mean difference [WMD] 5.34, 95 % CI: 3.85 to 6.8]; $p < 0.00001$) and plasma BNP level (WMD -117.37, 95 % CI: -227.22 to -7.52; $p = 0.04$) and no influence on overall mortality (RR 1.00, 95 % CI: 0.96 to 1.04; $p = 0.95$). The authors concluded that in the present meta-analysis, use of NPPV plus SMT improved LVEF and reduced plasma BNP level; but did not improve overall mortality in patients with CHF.

The authors noted that this meta-analysis had several drawbacks:

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I. the sample sizes of component trials included in this analysis were generally not large, which may bring “small-study effects”, which refer to the fact that trials with limited sample sizes were more likely to report larger beneficial effects than large trials,
II. only 2 studies included in this meta-analysis presented the data on NPPV compliance.

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II. only 2 studies included in this meta-analysis presented the data on NPPV compliance.

Ferrier et al pointed out patients using CPAP (greater than 1 hour per night CPAP) had the greatest increase in LVEF; Joho et al found that the change in average use of ASV correlated with changes in LVEF. However, the definition of NPPV compliance in those reports was not consistent and the influence of compliance to treatment was not quantified. Thus, the authors did not report the influence of NPPV compliance on studied variables in the present study.

Furthermore, an UpToDate review on “Overview of the therapy of heart failure with reduced ejection fraction” (Colucci, 2017) does not mention NPPV as a therapeutic option.

Gomes Neto and colleagues (2018) examined the effect of non-invasive ventilatory support (NIVS) with bi-level positive airway pressure ventilation (BiPAP) on exercise tolerance and dyspnea in HF patients. A total of 40 patients with NYHA class I/II/III HF were randomly assigned either to a NIVS group (n = 20) or control group (n = 20). All patients underwent two 6-min walk tests (6MWT), with a 30-min interval between them. In the NIVS group, the patients performed the BiPAP with an inspiratory positive airway pressure of 12 cmH2O and expiratory positive airway pressure of 6 cmH2O for 30 mins. At baseline, and after the 1st and 2nd 6MWT, the heart rate, systolic and diastolic blood pressure, peripheral oxygen saturation (SaO2), and dyspnea were evaluated. All 40 patients completed the study safely according to the randomization protocol, and no adverse events (AEs) were reported during the tests. The NIVS group showed a significant improvement in the 6MWT distance (68.3 versus 9.8 m) and dyspnea (1.3 versus 3.1) compared with the control group. No serious AEs were reported. The authors concluded that NIVS/BiPAP showed beneficial effects on exercise tolerance and dyspnea; it was safe and well-tolerated by HF patients and should be considered for inclusion in cardiac rehabilitation programs.

The authors stated that this study had several drawbacks. First, it was not possible to define the duration of the effects that were verified as the administration of the non-invasive ventilation was interrupted for the performance of the 6MWT. It would be important to determine whether several sessions with non-invasive ventilation would maintain the effect for a longer period of time and whether these effects are associated with clinical improvement. Furthermore, the lack of information about the hemodynamic profiles of the patients enrolled was another drawback.

NPPV for Asthma / Bronchiolitis in Infants and Children

Clayton and colleagues (2019) noted that initial respiratory support with NPPV or high-flow nasal cannula may prevent the need for invasive mechanical ventilation in pediatric intensive care unit (PICU) patients with bronchiolitis. However, it is unclear if the initial choice of respiratory support modality influences the need for subsequent invasive mechanical ventilation. These researchers compared the rate of subsequent invasive mechanical ventilation after initial support with NPPV or high-flow nasal cannula in children with bronchiolitis. This trial included a total of 92 participating PICUs. Children less than 2 years of age were admitted to a participating PICU between 2009 and 2015 with a diagnosis of bronchiolitis who were prescribed high-flow nasal cannula or NPPV as the initial respiratory treatment modality. Subsequent receipt of invasive mechanical ventilation was the primary outcome. These investigators identified 6,496 participants with a median age 3.9 months (1.7 to 9.5 months); most (59.7 %) were male, and 23.4 % had an identified co-morbidity. After initial support with NPPV or high-flow nasal cannula, 12.3 % of patients subsequently received invasive mechanical ventilation, which was more common in patients initially supported with NPPV compared with high-flow nasal cannula (20.1 % versus 11.0 %: $p < 0.001$). In a multi-

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variate logistic regression model that adjusted for age, weight, race, viral etiology, presence of a co-morbid diagnosis, and Pediatric Index of Mortality score, initial support with NPPV was associated with a higher odds of subsequent invasive mechanical ventilation compared with high-flow nasal cannula (OR, 1.53; 95 % CI: 1.24 to 1.88). The authors concluded that in this large, multi-center database study of infants with acute bronchiolitis that received initial respiratory support with high-flow nasal cannula or NPPV, use of the latter was associated with higher rates of invasive mechanical ventilation, even after adjusting for demographics, co-morbid condition, and severity of illness. These investigators stated that a large, prospective, multi-center trial is needed to confirm these findings.

Furthermore, UpToDate reviews on "Bronchiolitis in infants and children: Treatment, outcome, and prevention" (Piedra and Stark, 2019) and "Noninvasive ventilation for acute and impending respiratory failure in children" (Nagler and Cheifetz, 2019) do not mention NPPV as a therapeutic option.

Korang et al (2024) noted that asthma is one of the most common reasons for hospital admission among children, with significant economic burden and impact on QOL. Non-invasive positive pressure ventilation is increasingly used in the care of children with acute asthma, although the evidence supporting it is weak, and clinical guidelines do not offer any recommendations on its routine use. However, NPPV might be an effective way to improve outcomes for some children with asthma. A previous review did not show a clear benefit, but was limited by few studies with small sample sizes. This Cochrane review was an update of the previous review. These investigators examined the benefits and harms of NPPV as an add-on therapy to usual care (e.g., bronchodilators and corticosteroids) in children (less than 18 years of age) with acute asthma. They searched the Cochrane Airways Group Specialized Register, CENTRAL, Medline, and Embase. These researchers also carried out a search of ClinicalTrials.gov and the WHO ICTRP. They searched all data-bases from their inception to March 2023, with no restrictions on language of publication. These investigators included RCTs assessing NPPV as add-on therapy to usual care versus usual care for children hospitalized for acute asthma exacerbations. They included 3 RCTs that randomized 60 children with acute asthma to NPPV and 60 children to control. All included studies examined the effects of BiPAP for acute asthma in a PICU setting. None of the studies employed CPAP; the controls received standard care. The median age of children ranged from 3 to 6 years, and asthma severity ranged from moderate to severe. The primary outcome measures were all-cause mortality, serious AEs, and asthma symptom score. Secondary outcomes were non-serious AEs, HR- QOL, arterial blood gases and pH, pneumonia, cost, and length of stay (LOS) in PICU. None of the studies reported any deaths or serious AEs (except 1 study that reported intubation rate). Two trials reported asthma symptom score, each showing reductions in asthma symptoms in the BiPAP group. In 1 trial, the asthma symptom score was (MD -2.50, 95 % CI: -4.70 to -0.30, p = 0.03; 19 children) lower in the BiPAP group. In the other trial, a cross-over trial, BiPAP was associated with a lower mean asthma symptom score (MD -3.7; 16 children; very low certainty evidence) before cross-over; however, investigators did not report a standard deviation, and it could not be estimated from the 1st phase of the trial before cross-over. The reduction in both studies was above the pre-defined minimal important difference (MID). Overall, NPPV with standard care may reduce asthma symptom score compared to standard care alone; however, the evidence was very uncertain. The only reported serious AE was intubation rate in 1 trial, which had an intubation rate of 40 % and showed that BiPAP may result in a large reduction in intubation rate (RR 0.47, 95 % CI: 0.23 to 0.95; 78 children); however, the evidence was very uncertain. Post-hoc analysis revealed that BiPAP may result in a slight decrease in Los in PICU (MD -0.87 day, 95 % CI: -1.52 to -0.22; 100 children); however, the evidence was very uncertain. Meta-analysis or Trial Sequential Analysis was not possible because of insufficient reporting and different scoring systems. All 3 studies had high risk of bias with serious imprecision of results, resulting in very low certainty of evidence. The authors concluded that available evidence for NNPV is uncertain. NPPV may result in an improvement in asthma symptom score, decreased intubation rate, and slightly shorter Los in PICU; however, the evidence was of very low certainty. These researchers stated that larger RCTs with low risk of bias are needed.

Karmali et al (2025) stated that non-invasive assessments of lung volume distribution often require inhaled contrast and are limited by low regional resolution. These investigators examined a non-contrast imaging method of spatial lung volume displacement, adapted to evaluate changes with NIPPV. This trial examined regional lung volume displacement in 9 healthy volunteers (6 men, 3 women; age of 29 to 55 years; body mass index (BMI) of 20.2 to 31.3 kg/m2) using X-ray velocimetry (XV). Subjects were examined during

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tidal breathing and, also with 15 cm H2O inspiratory and 5 cm H2O expiratory pressures in a supine position. Regional specific ventilation (SV) was measured during tidal breathing and NIPPV. Mean specific ventilation (MSV, ml/ml), low volume region (LVR; % of less than 0.1 ml/ml), and high-volume region (HVR; % of greater than 0.3 ml/ml) were calculated as output variables. Images were segmented into lobar as well as central and peripheral zones; 2-way ANOVA and paired t-tests were used to determine regional differences within individuals and the effect of NIPPV. NIPPV increased MSV in both peripheral ($p = 0.01$) and central ($p = 0.02$) lung regions compared to tidal breathing. High volume regions increased in both peripheral ($p = 0.04$) and central regions ($p = 0.04$) during NIPPV. The authors concluded that the findings of this study showed that non-contrast imaging techniques could evaluate regional lung ventilation and re-distribution of lung volumes on NIPPV. Moreover, these researchers noted that heterogeneous responses to NIPPV may be associated with distinct distribution of ventilation, and further investigations are needed to determine differential response to NIPPV due to lung pathology among those with respiratory disease.

Prevention of Complications After Pulmonary Resection in Individuals with Lung Cancer

Torres and colleagues (2019) noted that pulmonary complications are often observed during the post-operative period following lung resection for patients with lung cancer. Some situations such as intubation, a long stay in the ICU, the high cost of antibiotics and mortality may be avoided with the prevention of post-operative pulmonary complications. Non-invasive positive pressure ventilation (NIPPV) is widely used in hospitals, and is thought to reduce the number of pulmonary complications and mortality after this type of surgery. In a systematic review, these researchers examined the benefits and harms of NIPPV for patients undergoing lung resection. This was an update of a Cochrane review first published in 2015. They evaluated the safety and effectiveness of NIPPV for preventing complications in patients following pulmonary resection for lung cancer. These investigators searched the Cochrane Central Register of Controlled Trials (CENTRAL), Medline, Embase, LILACS and PEDro until December 21, 2018, to identify potentially eligible trials. They did not use any date or language restrictions in the electronic searches. They searched the reference lists of relevant papers and contacted experts in the field for information about additional published and unpublished studies. These investigators also searched the Register of Controlled Trials (www.controlled-trials.com) and ClinicalTrials.gov (clinicaltrials.gov) to identify ongoing studies. They considered randomized or quasi-randomized clinical trials that compared NIPPV in the immediate post-operative period following pulmonary resection with no intervention or conventional respiratory therapy. Two authors collected data and examined trial risk of bias. Where possible, they pooled data from the individual studies using a fixed-effect model (quantitative synthesis), but where this was not possible they tabulated or presented the data in the main text (qualitative synthesis). Where substantial heterogeneity existed, they applied a random-effects model. Of the 190 references retrieved from the searches, 7 RCTs (1 identified with the new search) and 1 quasi-randomized trial fulfilled the eligibility criteria for this review, including a total of 486 patients; 5 studies described quantitative measures of pulmonary complications, with pooled data showing no difference between NIPPV compared with no intervention (RR 1.03; 95 % CI: 0.72 to 1.47); 3 studies reported intubation rates and there was no significant difference between the intervention and control groups (RR 0.55; 95 % CI: 0.25 to 1.20); 5 studies reported measures of mortality on completion of the intervention period. There was no statistical difference between the groups for this outcome (RR 0.60; 95 % CI: 0.24 to 1.53). Similar results were observed in the subgroup analysis considering ventilatory mode (bi-level versus CPAP). No study evaluated the post-operative use of antibiotics; 2 studies reported the length of ICU stay and there was no significant difference between the intervention and control groups (MD -0.75; 95 % CI: -3.93 to 2.43); 4 studies reported the length of hospital stay and there was no significant difference between the intervention and control groups (MD -0.12; 95 % CI: -6.15 to 5.90). None of the studies described any complications related to NIPPV. Of the 7 included studies, 4 were considered as “low risk of bias” in all domains, 2 were considered “high risk of bias” for the allocation concealment domain, and 1 was also considered “high risk of bias” for random sequence generation; 1 was considered “high risk of bias” for including subjects with more severe disease. The new study identified could not be included in the meta-analysis because its intervention differed from the other studies (use of pre- and post-operative NIPPV in the same population). The authors concluded that this review demonstrated that there was no additional benefit of using NIPPV in the post-operative period following pulmonary resection for all outcomes analyzed (pulmonary complications, rate of intubation, mortality, post-operative consumption of antibiotics, length of

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ICU stay, length of hospital stay and AEs related to NIPPV). However, the quality of evidence was “very low”, “low” and “moderate” since there were few studies, with small sample size and low frequency of outcomes. These researchers stated that new well-designed and well-conducted randomized trials are needed to answer the questions of this review with greater certainty.

Guidelines on enhanced recovery after lung surgery from the Enhanced Recovery After Surgery Society and the European Society of Thoracic Surgeons (Batchelor, et al., 2019) concluded: "Non-invasive positive pressure ventilation has been widely used to prevent atelectasis following lung surgery, but studies to date have failed to demonstrate any significant clinical benefits. . . . The routine use of postoperative non-invasive positive pressure ventilation cannot be recommended".

Exsufflation Belt

The Exsufflation Belt (Lung Assist), a wearable non-invasive device, is an intermittent, abdominal, daytime pressure ventilator. This device administers daytime abdominal pressure ventilation support for pulmonary restrictive or pulmonary obstructive breathing. It is intended to administer up to 18 breaths/min (BPM) and pressures up to 50 cm H20. It consists of a fabric corset that contains a specialized air/sac bladder, which is inflated by a patient-supplied self-cycling positive pressure ventilator. Positive pressure from the patient-supplied ventilator compresses the abdomen causing the diaphragm to rise resulting in active exhalation. When the bladder is deflated, natural and passive inhalation occurs. Benefits of the Exsufflation Belt, which are not available and not capable when using conventional positive pressure by a face-mask, mouth-piece, or artificial airway, may include improvement of quality of life (QOL), dignity and confidence in the workplace, eating in public settings, enhanced seating posture, ability to sit up for extended hours, decreased work of breathing with increased tidal volume, fully inflated lungs while resting of respiratory muscles during continued use, alternative to full-time use of face-mask or mouth-piece allowing the patient to speak and eat freely, reduced mid-facial hyperplasia and facial necrosis from long-term use of a mask, as well as aids in swallowing and coughing.

There is a lack of evidence regarding the effectiveness of the Exsufflation Belt for pulmonary restrictive or pulmonary obstructive breathing.

Multi-Function Home Ventilators

In April 2017, Ventilation, Oxygen, Cough, Suction, and Nebulizer (VOCSN) Unified Respiratory System (Ventec Life Systems, Inc.) received FDA 510(k) clearance to provide continuous or intermittent positive pressure ventilatory support for the care of individuals who require mechanical ventilation. The VOCSN is considered a Class II continuous ventilator device which may be used in invasive and non-invasive applications. VOCSN is intended for pediatric through adult patients weighing at least 5 kg. It is intended for use in home, hospital, institutional, and transport settings, including portable applications. VOCSN is a multifunction ventilator that offers mechanical ventilation, oxygen, cough, suction, and nebulizer therapies in one compact, portable, lightweight device. This respiratory system provides invasive, noninvasive, and mouthpiece ventilation and delivers a comprehensive set of modes and settings. The integral oxygen concentrator is intended for the administration of supplemental oxygen, delivering the equivalent of 6 L/min of oxygen or up to 40% FiO2 in adult patients. External high pressure and low pressure oxygen sources can be connected when needed. The integral cough assist option is intended for patients who are additionally unable to cough or clear secretions effectively, without changing the circuit. The device monitors patient breathing and triggers a cough at a natural point in the breathing cycle. The integral suction pump is intended for airway fluid removal and oral/pharyngeal hygiene. Hospital grade suction system provides consistent high flows throughout the entire suction experience. Patients remain connected to the ventilator at all times, without needing to disconnect circuits between uses. The system is designed to reduce the gaps in ventilation, decrease the risk of patient misconnection, and minimize exposure to the patient’s airway.

Breaths can be initiated (started) and cycled (ended) by either the VOCSN system or the patient. Depending on the ventilation mode and breath control settings chosen, VOCSN may provide the patient with mandatory, assist, or spontaneous breaths. VOCSN may be connected to a non-vented mask, tracheostomy or

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endotracheal tube to provide ventilation therapy using an active or passive Ventec One-CircuitTM. These circuit types include an exhalation valve designed to expel exhaled gases. VOCSN may be connected to a vented mask to provide ventilation therapy using the valveless Ventec One-CircuitTM. This circuit type does not include an exhalation valve, and is designed for use with a mask patient interface incorporating a fixed leak. Further, VOCSN may be connected to a nasal cannula to provide high-flow therapy (which is not yet available in the United States) using the valveless Ventec One-CircuitTM.

The multi-functional device is customizable and may include all five therapies and features, or just the set of therapies and features prescribed by the treating physician. Depending on its configuration, VOCSN will be able to provide Ventilation, +Oxygen Concentration, +Cough, +Suction, and/or +Nebulizer therapy. It may also be able to provide oxygen from an external high-pressure source and FiO2 monitoring (+Pro configurations) or low-pressure oxygen (all configurations). The configuration name is based on the therapies VOCSN can provide. For example, the V+C configuration provides Ventilation+Cough therapies. The configuration name is printed on the back of the device.

VOCSN, using a non-invasive interface, is contraindicated for any of the following conditions:

- Inability to maintain a patent airway
- Risk of aspirating gastric contents
- Epistaxis that causes pulmonary aspiration of blood
- Acute sinusitis or otitis media
- Hypotension

VOCSN, using Cough therapy, is contraindicated for any of the following conditions:

- A medical history of bullous emphysema
- Recent barotrauma
- Susceptible to pneumothorax or pneumo-mediastinum

VOCSN includes two removable, rechargeable batteries and an internal rechargeable battery, used to power the device when no external source of continuous power is available. Using the power supply, VOCSN can be connected to a wall outlet or other AC power source.

Long-Term Non-Invasive Ventilation in Children with Robin Sequence

Faily et al (2025) discussed the clinical characteristics and outcomes of children with Robin sequence (RS) using NIV, including CPAP and BiPAP, and compared these parameters to other children using this technology. This study was a sub-study of a retrospective, multi-center, 10-year cohort of children using long-term NIV. Children with RS were identified from medical chart review and matched by age, sex, and year of initiation to other children in the cohort. Clinical characteristics, NIV technology, and treatment effect and outcomes were extracted from the medical chart. From 622 children in the NIV cohort, 13 had RS and were matched to 39 comparators. Age at NIV initiation and co-morbidities did not differ between the 2 groups. Use of gastrostomy/naso-gastric (NG) tubes was higher in children with RS (OR 3.0, 95 % CI: 1.17 to 7.69). Neither the proportion of children with OSA or obstructive AHI of 10 events/hour or higher differed between groups. Improvements in respiratory, oxygen, and CO2 parameters between the diagnostic and treatment PSG were similar for children with RS and the comparison group. NIV start location was predominantly at home and for used during sleep in both groups. Children with RS used NIV for a median of 1.45 (inter-quartile range [IQR] of 1.85) years. The most common reason for stopping NIV in both groups was an improvement in the underlying condition. The authors concluded that children with RS exhibited similar characteristics and outcomes to other children using NIV. A high proportion of children with RS discontinued NIV use because of improvements in the underlying primary condition resulting in NIV.

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Glossary of Terms

Term	Definition
Apnea	Cessation of airflow for at least 10 seconds
Apnea-hypopnea index (AHI)	Average number of episodes of apnea and hypopnea per hour of sleep without the use of a positive airway pressure device
A bilevel positive airway pressure (PAP) device *without* backup rate	Delivers adjustable, variable levels (within a single respiratory cycle) of positive air pressure by way of tubing and a noninvasive interface (such as a nasal, oral, or facial mask) to assist spontaneous respiratory efforts and supplement the volume of inspired air into the lungs
A bilevel PAP device *with* backup rate	Delivers adjustable, variable levels (within a single respiratory cycle) of positive air pressure by way of tubing and a noninvasive interface (such as a nasal or oral facial mask) to assist spontaneous respiratory efforts and supplement the volume of inspired air into the lungs. In addition, it has a timed backup feature to deliver this air pressure whenever sufficient spontaneous inspiratory efforts fail to occur.
Central apnea-hypopnea index (CAHI)	For diagnosis of central sleep apnea (CSA) the central apnea-central hypopnea index (CAHI) is defined as the average number of episodes of central apnea and central hypopnea per hour of sleep without the use of a positive airway pressure device. For complex sleep apnea (CompSA), the CAHI is determined during the use of a positive airway pressure device after obstructive events have disappeared.
Central sleep apnea (CSA)	I. An AHI greater than 5; and II. The sum total of central apneas plus central hypopneas is greater than 50% of the total apneas and hypopneas; and III. A central apnea-central hypopnea index (CAHI) is greater than or equal to 5 per hour; and IV. The presence of one or more of the following: A. Sleepiness B. Difficulty initiating or maintaining sleep, frequent awakenings, or non-restorative sleep C. Awakening short of breath D. Snoring E. Witnessed apneas; and V. There is no evidence of daytime or nocturnal hypoventilation.
Complex sleep apnea (CompSA)	A form of central apnea specifically identified by the persistence or emergence of central apneas or hypopneas upon exposure to CPAP or a bilevel PAP device without a backup rate feature when obstructive events have disappeared. These individuals have predominantly obstructive or mixed apneas during the diagnostic

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	sleep study occurring at greater than or equal to 5 times per hour. With use of CPAP or bilevel PAP without a backup rate feature, they show a pattern of apneas and hypopneas that meets the definition of CSA described above. Complex sleep apnea (CompSA) is a form of central apnea specifically identified by all of the following: I. With use of a positive airway pressure device without a backup rate, the polysomnogram (PSG) shows a pattern of apneas and hypopneas that demonstrates the persistence or emergence of central apneas or central hypopneas upon exposure to CPAP or a bi-level device without backup rate device when titrated to the point where obstructive events have been effectively treated (obstructive AHI less than 5 per hour); *and* II. After resolution of the obstructive events, the sum total of central apneas plus central hypopneas is greater than 50% of the total apneas and hypopneas; *and* III. After resolution of the obstructive events, a central apnea-central hypopnea index (CAHI) greater than or equal to 5 per hour.
FEV1	Forced expiratory volume in 1 second
FIO2	Fractional concentration of oxygen delivered to the member for inspiration. The member's usual FIO2 refers to the oxygen concentration the member normally breathes when not undergoing testing to qualify for coverage of NPPV. That is, if the member does not normally use supplemental oxygen, their usual FIO2 is that found in room air.
FVC	Forced vital capacity
Hypopnea	Abnormal respiratory event lasting at least 10 seconds associated with at least a 30 % reduction in thoraco-abdominal movement or airflow as compared to baseline, and with at least a 4 % decrease in oxygen saturation.
Polysomnography	Continuous and simultaneous monitoring and recording of various physiological and pathophysiological parameters of sleep with physician review, interpretation, and report.
Respiratory cycle	Inspiration, followed by an expiration
Type I sleep test	Continuous and simultaneous monitoring and recording of various physiological and pathophysiological parameters of sleep with physician review, interpretation, and report.
Type II device	Monitors and records a minimum of seven (7) channels: EEG, EOG, EMG, ECG/heart rate, airflow, respiratory movement/effort and oxygen saturation
Type III device	Monitors and records a minimum of four (4) channels: respiratory movement/effort, airflow, ECG/heart rate and oxygen saturation
Type IV device	Monitors and records a minimum of three (3) channels, one of which is airflow

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Appendix

Apnea-hypopnea index (AHI): If the AHI is calculated based on less than 2 hours of continuous recorded sleep, the total number of recorded events used to calculate the AHI must be at least the number of events that would have been required in a 2-hour period (i.e., greater than or equal to 10 events).

Central apnea-hypopnea index (CAHI) - For diagnosis of central sleep apnea (CSA), if the CAHI is calculated based on less than 2 hours of continuous recorded sleep, the total number of recorded events used to calculate the CAHI must be at least the number of events that would have been required in a 2-hour period (i.e., greater than or equal to 10 events).

A polysomnography must include sleep staging, which is defined to include a 1 to 4 lead electroencephalogram (EEG), an electrooculogram (EOG), and a submental electromyogram (EMG). It must also include at least the following additional parameters of sleep: airflow, respiratory effort, and oxygen saturation by oximetry. It may be performed either as a whole-night study for diagnosis only or as a split-night study to diagnose and initially evaluate treatment. For indications other than OSA, polysomnography studies must be performed in a sleep study laboratory, and not in the home or in a mobile facility. According to DME MAC policy (NHIC, 2008), arterial blood gas, sleep oximetry and polysomnographic studies may not be performed by the DME supplier.

Note on Sleep Testing

Payment for a bilevel PAP device for the treatment of the conditions specified in this policy may be contingent upon an evaluation for the diagnosis of sleep apnea (obstructive sleep apnea (OSA), central sleep apnea (CSA), or complex sleep apnea (CompSA). The sleep test must be either a polysomnogram performed in a facility-based laboratory (Type I study) or an inpatient hospital-based or home based sleep test (HST) (Types II, III, IV, Other).

A type I sleep test is facility-based and must include sleep staging, which is defined to include a 1-4 lead electroencephalogram (EEG), electro-oculogram (EOG), submental electromyogram (EMG) and electrocardiogram (ECG). It must also include at least the following additional parameters of sleep: airflow, respiratory effort, and oxygen saturation by oximetry. It may be performed as either a whole night study for diagnosis only or as a split night study to diagnose and initially evaluate treatment.

Not all types of HST are appropriate for the evaluation of CSA or CompSA, as they do not monitor the necessary parameters. An HST is performed unattended in the member's home or during a hospitalization using a portable monitoring device. A portable monitoring device for conducting an HST must meet one of the following criteria:

- Type II device; *or*
- Type III device; *or*
- Type IV device; *or*
- Other - devices that monitor and record a minimum of three (3) channels that include actigraphy, oximetry and peripheral arterial tone and for which there is substantive clinical evidence in the published peer-reviewed medical literature that demonstrates that the results accurately and reliably correspond to an AHI or RDI as defined above. This determination will be made on a device-by-device basis.

Table: Usual Medically Necessary Quantities of Supplies for Use With a NPPV:

Supply	Quantity Usually Medically Necessary
Tubing with integrated heating element	1 per 3 months

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Combination oral/nasal mask	1 per 3 months
Oral cushion for combination oral/nasal mask	2 per 1 month
Nasal pillows for combination oral/nasal mask (pair)	2 per 1 month
Full face mask	1 per 3 months
Replacement face mask interface for full face mask	1 per 1 month
Cushion for use on nasal mask interface	2 per 1 month
Pillow for use on nasal cannula type interface (pair)	2 per 1 month
Nasal interface (mask or cannula type), with or without head strap	1 per 3 months
Headgear	1 per 6 months
Chinstrap	1 per 6 months
Tubing	1 per 3 months
Filter (disposable)	2 per 1 month
Filter (nondisposable)	1 per 6 months
Water chamber with humidifier	1 per 6 months

Source: NHIC, 2008.

Tubing with integrated heating element for use with a positive airway pressure device describes tubing used with a heated humidifier and has a heated wire running the length of the tubing. It is designed for use with a positive airway pressure device and a non-invasive interface - i.e., nasal or face mask, nasal cannula, or oral interface.

Cushion for use on nasal mask interface, replacement only, each is used for a replacement nasal mask interface that goes around the nose, but not into the nostrils. The unit of service for this code is “each”.

Pillow for use on nasal cannula type interface, replacement only, pair is used for a replacement nasal cannula-type interface. This interface extends a short distance into the nostrils. The unit of service for this code is “pair”. For some products, there are two physically separate cushions or “pillows” – one for each nostril. Two cushions/ pillows equal one unit of service. For other products, the interface is a single piece with two protrusions that extend into the nostrils. One of these interfaces equals one unit of service.

A combination oral/nasal mask, used with continuous positive airway pressure device, each, is a two piece system with separate elements for oral and nasal use.

A liner is a soft, flexible material, which is placed between the member’s skin and the PAP mask interface. Liners used with a PAP mask are made of cloth, silicone or other materials. Liners are not interfaces for use with a PAP mask. Consequently, liners should not be billed as replacement features of a PAP mask.

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The above policy is based on the following references:

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